

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

MOTION IN STRAIGHT LINE

LECTURE-2

PHYSICS

WAJAHID SIR (Gabbar



Topics To Be Covered

- 1 Revision ✓
I - Speed | A - speed | I - velocity | a - velocity
- 2 Acceleration
- 3 Positive Acceleration
- 4 Negative Acceleration
- 5 Zero Acceleration

Graph ✓

Lecture no. 2

Speed

→ Scalar
direction x

$$\underline{143 \div 3}$$

Average Speed

Instantaneous Speed

Neet

$$\frac{\text{Average}}{\text{Speed}} = \frac{\text{Total distance}}{\text{Total time}}$$

→ Average
Speed = $\frac{\Delta x}{\Delta t}$

Neet

$$\frac{\text{Instantaneous}}{\text{Speed}} = \frac{\text{Small distance cover}}{\text{Small time taken}}$$

Instantaneous
Speed = $\frac{dx}{dt}$

Q. (1) Numerical



Average speed = $\frac{\text{Total distance}}{\text{Total time}}$

$$x_2 = x_1 + T_d$$

$$\boxed{x_2 - x_1 = T_d}$$

$$= \frac{100\text{m}}{10\text{s}}$$

$$= 10\text{m/s}$$

delta $\rightarrow \frac{\Delta x}{\Delta t}$

$$\text{Average speed} = \frac{T_d}{T_{\text{time}}}$$

$$= \frac{x_2 - x_1}{t_1}$$

$$t_2 \rightarrow \frac{4:00 - 4:00}{10\text{sec}}$$

$$= \frac{x_2 - x_1}{t_2 - t_1} = \frac{\Delta x}{\Delta t} \checkmark$$



Sol Average speed = $\frac{Td}{Tt} = \frac{200}{50} = 4 \text{ m/s}$

$$x_2 = x_1 + Td$$

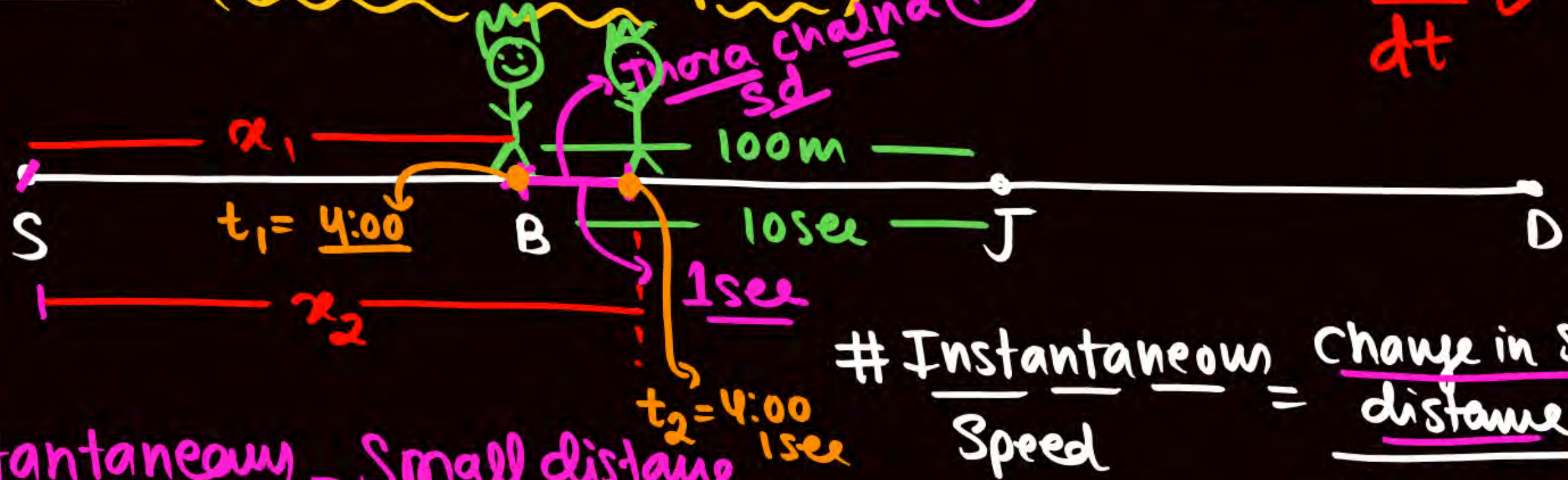
$$x_2 - x_1 = Td$$

$$= \frac{x_2 - x_1}{t_2 - t_1}$$

$$= \frac{210 - 10}{8:00 - 8:00 \text{ 50sec}} = \frac{200}{50} = 4 \text{ m/s}$$

Q Numerical (Instantaneous Speed)

$$IS = \frac{dx}{dt} \checkmark$$



$$\# \text{ Instantaneous Speed} = \frac{\text{Small distance}}{\text{Small time}}$$

$$\# x_2 = x_1 + Sd$$

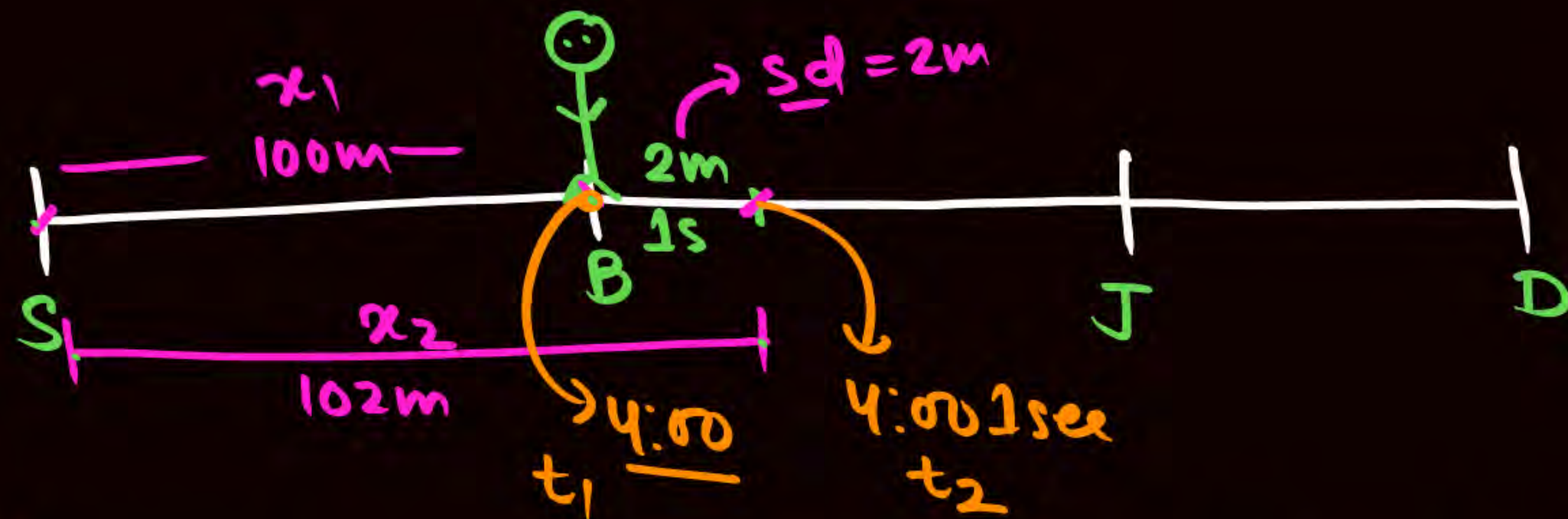
$$\boxed{x_2 - x_1 = Sd}$$

$$= \frac{1m}{1s} = 1m/s$$

$$\# \text{ Instantaneous Speed} = \frac{\text{Change in Small distance}}{\text{Change in Small time}}$$

$$\begin{aligned} \text{Small change} & \rightarrow x_2 - x_1 \\ & = \frac{x_2 - x_1}{t_2 - t_1} \\ & = \frac{dx}{dt} \checkmark \end{aligned}$$

Q.



Sol Instantaneous Speed = $\frac{\text{Small change in distance}}{\text{Small change in time}} = \frac{2m}{1} = 2m/s$

$$= \frac{dx}{dt}$$

$$= \frac{x_2 - x_1}{t_2 - t_1} = \frac{102 - 100}{4:00 - 4:00.1 \text{ sec}} = \frac{2}{1} = 2m/s$$

Q. the distance as the function of (Neet-2017)

time is given by $x = t^3 + t^2 + t + 2$

Calculate the instantaneous speed = ? at $t = \underline{1\text{ sec}}$ ✓ 1 sec ✓ 1 sec

Sol

$$x = t^3 + t^2 + t + 2$$

$$I_s = \frac{dx}{dt} = \frac{d}{dt}(t^3 + t^2 + t + 2)$$

$$I\text{-Speed} = \frac{d}{dt}t^3 + \frac{d}{dt}t^2 + \frac{d}{dt}t + \frac{d}{dt}2$$

$$= 3t^{3-1} + 2t^{2-1} + 1t^{1-1} + 0$$

$$= 3t^2 + 2t + 1 \times t^0$$

$$= 3t^2 + 2t + 1 \times 1$$

$$I_s = 3t^2 + 2t + 1$$

$$I_s = 3 \times 1^2 + 2 \times 1 + 1$$

$$= 3 \times 1 + 2 \times 1 + 1$$

$$= 3 + 2 + 1$$

$$= 5\text{ m/s} \checkmark$$

Velocity

Vector
direction ✓

Average velocity

Instantaneous velocity

Need

$$\text{Average} = \frac{\text{Total displacement}}{\text{Total time}}$$

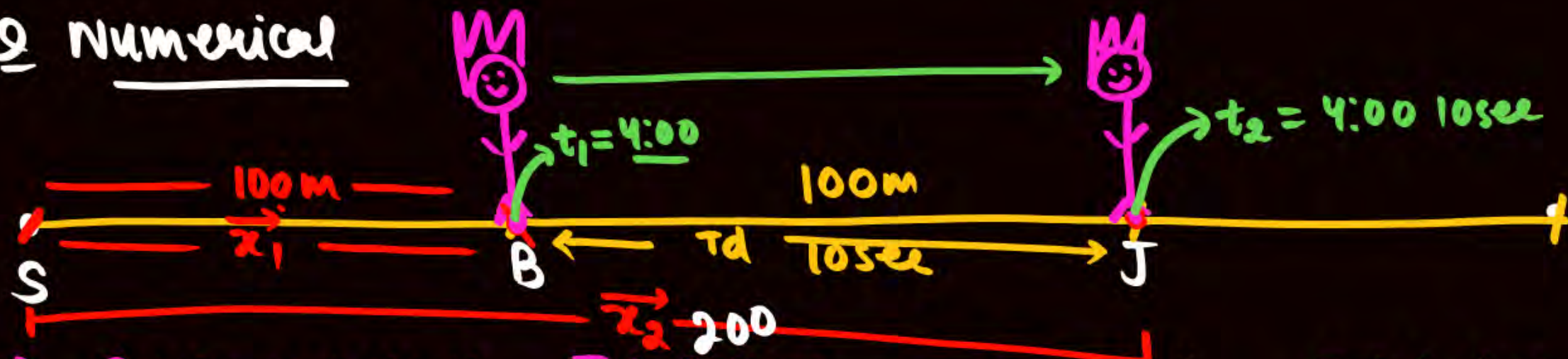
→ $\text{Average Velocity} = \frac{\Delta \vec{x}}{\Delta t}$

Need

$$\text{Instantaneous Speed} = \frac{\text{Small displacement}}{\text{Small time taken}}$$

→ $\text{Instantaneous Velocity} = \frac{d\vec{x}}{dt}$

Q Numerical



Sol Average Velocity = $\frac{\text{Total displacement}}{\text{Total time}}$

$$\# \vec{x}_2 = \vec{x}_1 + \vec{Td}$$

$$\vec{x}_2 - \vec{x}_1 = \vec{Td}$$

$$= \frac{100\text{m}}{10\text{s}}$$

$$= 10\text{m/s} (\rightarrow)$$

Right

Average =
Velocity

Total change in
displacement
Total change in time

$$\text{Average Velocity} = \frac{\vec{Td}}{\vec{T}\text{time}}$$

$$= \frac{\vec{x}_2 - \vec{x}_1}{t_2 - t_1} = \frac{\Delta \vec{x}}{\Delta t}$$

delta
change

$$\text{Average Velocity} = \frac{\Delta \vec{x}}{\Delta t}$$

$$= \frac{\vec{x}_2 - \vec{x}_1}{t_2 - t_1}$$

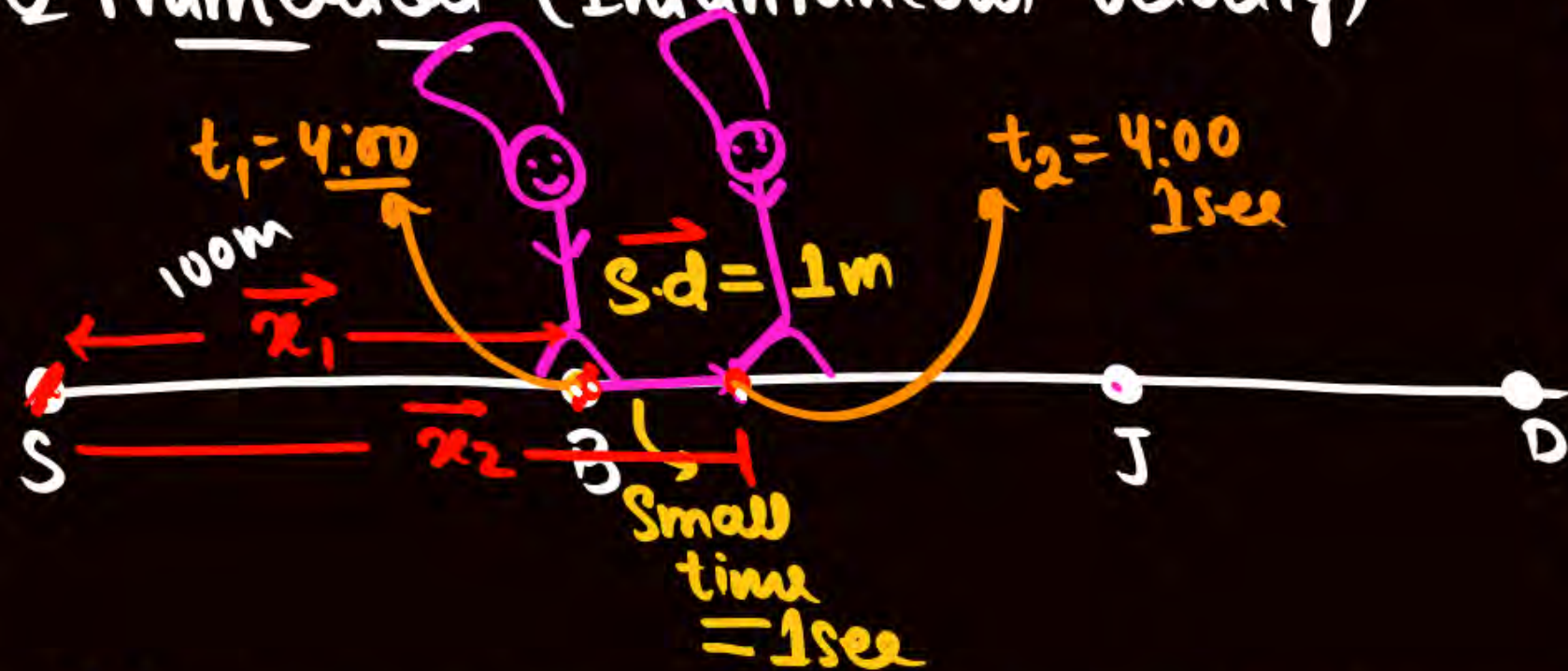
$$= \frac{200 - 100}{4:00 10\text{sec} - 4:00}$$

$$= \frac{100}{10}$$

$$= 10\text{m/s} (\rightarrow)$$

Right

Q Numerical (Instantaneous Velocity)



Sol Instantaneous Velocity = $\frac{\text{Small displacement}}{\text{Small time}}$

$x_2 = x_1 + x_d$
 $\vec{x}_2 - \vec{x}_1 = \vec{x}_d$

$= \frac{1m}{1s} = \underline{\underline{1m/s}} \rightarrow \underline{\underline{\text{Right}}}$

$I.V = \frac{\text{Small change in } \vec{x}}{\text{Small change in } t}$
 $= \frac{\vec{x}_2 - \vec{x}_1}{t_2 - t_1} \rightarrow \text{small}$

$I.V = \frac{d\vec{x}}{dt} \checkmark$

$I.V = \frac{d\vec{x}}{dt} = \frac{\vec{x}_2 - \vec{x}_1}{t_2 - t_1}$

$= \frac{101 - 100}{4:00 \text{ 1sec} - 4:00} = \frac{1}{1s} = 1m/s \rightarrow \text{Right}$

Q If the displacement $\vec{x}$ as the function of time $\uparrow$ Instantaneous is given by $(t^2 + t + 2)$ Find out the velocity at $t = 2 \text{ sec}$? (neet - 2023)

Sol

$$V = \frac{d\vec{x}}{dt}$$

$$V = \frac{d(t^2 + t + 2)}{dt}$$

$$V = \frac{dt^2}{dt} + \frac{dt^1}{dt} + \frac{d2}{dt}$$

$$V = 2t^{2-1} + 1t^{1-1} +$$

$$V = 2t^1 + 1 \times t^0$$

$$V = 2t + 1 \times 1$$

$$V = 2t + 1 \quad (t = 2 \text{ sec})$$

$$V = 2(2) + 1 = 5 \text{ m/s}$$

(Right)

Q
4/5 $x = 4t^3 + 2t^2 + 2t + 5$ $\rightarrow$ Speed at $t = 2\text{ sec}$?

SUMMARY



- Speed $\begin{cases} \rightarrow AS = \frac{\Delta x}{\Delta t} \\ \rightarrow IS = \frac{dx}{dt} \end{cases}$
- Velocity $\begin{cases} \rightarrow AV = \frac{\Delta \vec{x}}{\Delta t} \\ \rightarrow IV = \frac{d\vec{x}}{dt} \end{cases}$

Thank
You